In the table of 100 simulated sets, in what proportion were there 7 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

If a model is consistent with the data, the simulation proves that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

Decide whether a model is consistent with data

MAIN	Probability & Statistics main question	**HSS-IC.A.2**
BACKWARD 1	Read the results of a simulation	**HSS-IC.A.2**
BACKWARD 2	Compare observed frequencies with a probability model	**7.SP.C.7.b**
FORWARD 1	Decide using results extreme in either direction	**HSS-IC.A.2**
FORWARD 2	Judge a treatment difference by re-randomization	**HSS-IC.B.5**

SET 24 • HSS-IC.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

The table shows 100 simulated sets of 10 fair-coin flips. In what proportion of the sets were there 8 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

A coin claimed to be fair gave 8 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: the model is a fair coin. Count the simulated results that are 8 or more heads (a one-sided check: only an excess of heads counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin should give 5 heads every time.

B. Consistent: 8 heads happened in exactly 4% of the sets.

C. Not consistent: 8 or more heads happened in only 6% of the sets.

D. Consistent: 8 or more heads happened in 6% of the simulated sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5

(Hypothetical simulation results.) A die is claimed to be fair. In 60 rolls it showed 18 sixes. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. What proportion of the simulated sets is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

(Hypothetical simulation results.) A die claimed to be fair showed 18 sixes in 60 rolls. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. Decision rule: the model is a fair die. Count the simulated results that are 18 or more sixes (a one-sided check: only an excess of sixes counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: 18 sixes is possible with a fair die.

B. Not consistent: so many sixes happened in only 2% of simulated sets.

C. Not consistent: a fair die must give exactly 10 sixes.

D. Consistent: 4 simulated sets had 18 or more sixes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5

(Hypothetical simulation results.) A spinner is claimed to land on red 25% of the time. In 40 spins it landed on red 6 times. In 1,000 simulated sets of 40 spins, 6 or fewer reds happened 120 times. Decision rule: the model is a spinner that lands on red 25% of the time. Count the simulated results that are 6 or fewer reds (a one-sided check: only too few reds counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: the spinner landed on red exactly 25% of the time.

B. Consistent: 6 or fewer reds happened in 12% of the sets.

C. Not consistent: only 120 of the sets had 6 or fewer.

D. Not consistent: 6 is fewer than the expected 10 reds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 1

Read the results of a simulation • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 50 simulated families of 4 children, how many families had exactly 2 girls?

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Families: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

In the table of 50 simulated families, what proportion had 1 or fewer girls? Give a decimal.

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

True or false?

The table of 100 simulated sets of 10 fair-coin flips shows 25 sets with exactly 5 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

In the table of 100 simulated sets of 10 fair-coin flips, how many sets had 3 or fewer heads?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. 12

B. 17

C. 5

D. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

True or false?

In the table of 100 simulated sets of 10 fair-coin flips, more sets had 6 heads than 4 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 2

Compare observed frequencies with a probability model • Grade 7 • Statistics & Probability • 7.SP.C.7.b

True or false?

A coin that lands heads 520 times in 1,000 flips is reasonably consistent with a fair coin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5

A fair die is rolled 600 times, independently. About how many sixes does the fair model predict?

A. 160

B. 60

C. 100

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5

How many heads does a fair-coin model predict on average in 10 flips?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5

True or false?

A fair die rolled 60 times, independently, must show each face exactly 10 times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5

A spinner is claimed to land on red 50% of the time. Which result in 100 spins most suggests the claim is wrong?

A. 82 red

B. 50 red

C. 47 red

D. 53 red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 1

Decide using results extreme in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 100 simulated sets of 10 fair flips, what proportion had 9 or more heads or 1 or fewer heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

A coin claimed to be fair gave 1 head in 10 flips. Use the table of 100 simulated sets of 10 fair-coin flips; results at least as far from 5 heads as 1 head are in either tail. Decision rule: the model is a fair coin. Count the simulated results that are 9 or more heads or 1 or fewer heads (a two-sided check: too many or too few heads both count). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin cannot give 1 head.

B. Not consistent: such results happened in 3% of the sets.

C. Consistent: 1 head happened in exactly 1 set.

D. Consistent: such results happened in 3% of the sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

True or false?

Under the 5% rule, the smaller the proportion of simulated results as extreme as the data, the stronger the evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

A coin claimed to be fair gave 7 heads in 10 flips. In the table, what proportion of sets are at least as far from 5 heads in either direction (7 or more, or 3 or fewer)? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

True or false?

If a model is judged not consistent under the 5% rule, there is no chance that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 2

Judge a treatment difference by re-randomization • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

(Hypothetical simulation results.) In 100 re-randomizations of an experiment's results, a difference in means of 6 or more appeared 3 times. What proportion is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

(Hypothetical simulation results.) In a randomized experiment, the treatment group's mean minus the control group's mean was 6. In 100 re-randomizations of the results, differences (treatment minus control) of 6 or more appeared 3 times. Decision rule: the model is the "no effect" model (the treatment does not change the results). Count the simulated results that are differences of 6 or more (a one-sided check). If they make up fewer than 5% of the simulated results, the "no effect" model is not consistent with the data and the difference is significant; otherwise it is consistent, and the difference is not significant. Which decision is correct?

A. The difference is significant: every difference is significant.

B. The difference is not significant: it appeared 3 times.

C. The difference is significant: chance alone gave it 3% of the time.

D. The difference is not significant: re-randomizing changes the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5

True or false?

A re-randomization simulation assumes that the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

In an experiment, the treatment group's mean was 24 and the control group's mean was 19. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

True or false?

If the observed difference appeared in 40% of re-randomizations, the treatment effect is significant under the 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

SET 25 • HSS-IC.B.3 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Sample surveys, experiments and observational studies

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Identify the population, sample, parameter and statistic	**HSS-IC.A.1**
BACKWARD 2	Separate association from causation	**HSS-ID.C.9**
FORWARD 1	Identify the parts of an experiment	**HSS-IC.B.3**
FORWARD 2	Choose a study design for a question	**HSS-IC.B.3**

SET 25 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A researcher randomly assigns 60 volunteers to a new sleep routine or their usual routine and compares their test scores. What kind of study is this?

A. census

B. sample survey

C. experiment

D. observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5

A polling company asks a random sample of 1,000 adults which candidate they prefer. What kind of study is this?

A. experiment

B. simulation

C. sample survey

D. observational study comparing groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5

Researchers compare lung health of people who already smoke with people who do not. What kind of study is this?

A. experiment

B. census

C. observational study

D. sample survey of opinions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5

Which kind of study is designed mainly to estimate a characteristic of a population?

A. an observational study of two groups

B. a sample survey

C. a simulation of a model

D. an experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5

Why can an experiment support a cause-and-effect conclusion when an observational study cannot?

A. The researcher imposes treatments, so other variables can be controlled.

B. Observational studies never collect any numerical data.

C. Experiments do not need any comparison or control group.

D. Experiments always use more subjects than observational studies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 1

Identify the population, sample, parameter and statistic • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

300 randomly chosen city residents are asked how often they ride the bus. What is the population?

A. the 300 residents

B. all city residents

C. the city's buses

D. bus riders only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5

300 randomly chosen city residents are asked how often they ride the bus. What is the sample size?

Size: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

A statistic describes a sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A census collects data from only part of the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

Which is a parameter?

A. the mean height of 50 sampled students

B. the number 50

C. the proportion in the sample

D. the mean height of all students in a state

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 2

Separate association from causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

True or false?

An association seen without assigning treatments may be caused by a lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

Towns with more ice-cream shops have more drownings. What is a likely lurking variable?

A. hot weather

B. the number of shops

C. the price of ice cream

D. the town's name

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

A correlation always implies causation when the study is large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

Which conclusion requires an experiment?

A. The mean recovery time was 6 days.

B. Recovery time and vitamin use are associated.

C. Half of the patients took the vitamin.

D. A vitamin causes faster recovery.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

Random assignment helps rule out lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 1

Identify the parts of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

An experiment compares two fertilizers on 40 plots and measures crop yield. What is the response variable?

A. the 40 plots

B. the two fertilizers

C. the farmer

D. crop yield

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5

An experiment tests 3 doses of a medicine and a placebo. How many treatments are there?

Treatments: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

True or false?

A control group receives the standard treatment or a placebo so the new treatment can be compared.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

Why do experiments on medicines often give some subjects a placebo?

A. to increase the number of different treatments

B. to make the experiment cheaper to run

C. to account for the effect of believing one is being treated

D. to make sure the results come out positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

In a double-blind experiment, only the subjects do not know which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 2

Choose a study design for a question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Question: does a new math app raise test scores? Which design fits best?

A. a randomized experiment

B. a census of all students

C. a sample survey

D. an observational study of app users

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5

Question: what percent of a town's residents own a bicycle? Which design fits best?

A. a randomized experiment

B. a sample survey

C. a simulation

D. an observational study of cyclists

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

Question: do people who live near highways have more asthma? Assigning where people live is not possible. Which design fits best?

A. an observational study

B. a sample survey of opinions

C. a simulation

D. a randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

It is ethical to randomly assign people to smoke in an experiment on lung disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

An observational study can be the best choice when assigning a treatment would be unethical.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

How randomization relates to each kind of study

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Classify a study	**HSS-IC.B.3**
BACKWARD 2	Select a simple random sample	**7.SP.A.1**
FORWARD 1	Decide the scope of conclusions	**HSS-IC.B.3**
FORWARD 2	Evaluate a report's conclusion from its design	**HSS-IC.B.6**

SET 26 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A class has 40 students numbered 01 to 40. Read line 1 of the table as two-digit numbers, skipping 00, numbers above 40 and repeats. Which three students are chosen first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5

What is the main purpose of selecting a sample at random in a survey?

A. so the sample is as small as possible

B. so that every answer comes out the same

C. so a treatment can be shown to cause a change

D. so the results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5

What is the main purpose of assigning subjects to treatments at random in an experiment?

A. to guarantee that the treatment will work

B. to help balance other variables between the groups

C. to avoid needing any control or comparison group

D. to let the results describe everyone in the country

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5

A study randomly assigned 80 volunteers to two diets and found a significant difference in weight loss. The volunteers were not a random sample. What can be concluded?

A. The diet causes the same difference for everyone in the country.

B. The diet caused the difference for volunteers like these.

C. Nothing, because the subjects were volunteers.

D. Only that the two diets are associated with weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5

Plots in a field are numbered 01 to 60. Read line 3 of the table as two-digit numbers, skipping 00, numbers above 60 and repeats. Which three plots are assigned fertilizer A first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 1

Classify a study • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A teacher asks a random sample of 50 students how many hours they sleep. What kind of study is this?

A. observational study comparing treatments

B. sample survey

C. census

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5

True or false?

In an experiment, the researcher decides which subjects get which treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

Doctors compare the heart health of people who already jog with people who do not. What kind of study is this?

A. sample survey

B. simulation

C. observational study

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

A sample survey assigns treatments to people.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

Which study imposes a treatment?

A. giving half of the volunteers a new drug, chosen by lottery

B. asking a random sample of voters for their opinion

C. counting the cars that pass along a busy road

D. recording the diets that people already choose to eat

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 2

Select a simple random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

True or false?

In a simple random sample, every group of the same size has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5

Which method gives a simple random sample of 5 students from a class of 30?

A. draw 5 names from a hat holding all 30 names

B. ask for 5 volunteers to raise their hands

C. choose the 5 tallest students in the class

D. choose the first 5 students to arrive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5

Students are numbered 01 to 30. Reading the two-digit numbers 45 07 29 07 33 18, skipping numbers above 30 and repeats, which three students are chosen?

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5

True or false?

Choosing every student whose name starts with A is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5

Why are repeated numbers skipped when selecting a sample from a random-digit table?

A. skipping them saves time in the selection

B. repeated digits are always printing errors

C. repeated numbers are too large to use

D. a person cannot be chosen twice

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 1

Decide the scope of conclusions • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A study used a random sample from a population but no random assignment. What can be concluded?

A. Nothing at all can be concluded from such a study.

B. Associations generalize to the population, but not cause.

C. Cause can be concluded, but only for the sample.

D. Cause can be concluded for the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5

A study used both a random sample from a population and random assignment to treatments. What can be concluded?

A. Cause, but only for the sample.

B. Association only.

C. Cause, generalized to the population.

D. Nothing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5

True or false?

Random assignment allows cause-and-effect conclusions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

True or false?

Random sampling allows cause-and-effect conclusions even without random assignment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

A study used neither random sampling nor random assignment. What can be concluded?

A. Cause for the group studied.

B. Cause for the population.

C. Associations for the population.

D. Associations in the group studied only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 2

Evaluate a report's conclusion from its design • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news report says, "People who drink green tea live longer, so drink tea to live longer." The data came from an observational study. What is the main flaw?

A. The sample was too large to be trusted.

B. Green tea intake cannot be measured at all.

C. Observational studies cannot find associations.

D. It claims cause from an observational study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5

True or false?

A report based on a voluntary online poll can be generalized to all adults.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

A randomized experiment on 200 college athletes found a new drink improved sprint times. To whom does the result apply most directly?

A. all adults

B. college athletes like those studied

C. all children

D. nobody

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5

True or false?

A report should state how its sample was chosen and whether treatments were randomly assigned.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

A survey of 50 gym members found 90% exercise daily, reported as "90% of Americans exercise daily." What is wrong?

A. Surveys can never be used to estimate percents.

B. Gym members are not representative of all Americans.

C. The sample of 50 people is far too large.

D. The percent of 90 was computed incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5

Estimate a population proportion with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Compute a sample proportion and name it	**HSS-IC.A.1**
BACKWARD 2	Read a standard deviation as typical variation	**HSS-ID.A.2**
FORWARD 1	See how sample size changes the margin of error	**HSS-IC.B.4**
FORWARD 2	Read margins of error in poll reports	**HSS-IC.B.6**

SET 27 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

In a random sample of 400 adults, 248 favor a new park. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

(Hypothetical simulation results.) 500 random samples of size 400 were simulated from a population with proportion 0.62. The simulated sample proportions had standard deviation 0.024. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

A sample proportion is 0.62 and the margin of error is 0.048. Give the interval of plausible values for the population proportion.

From: _____

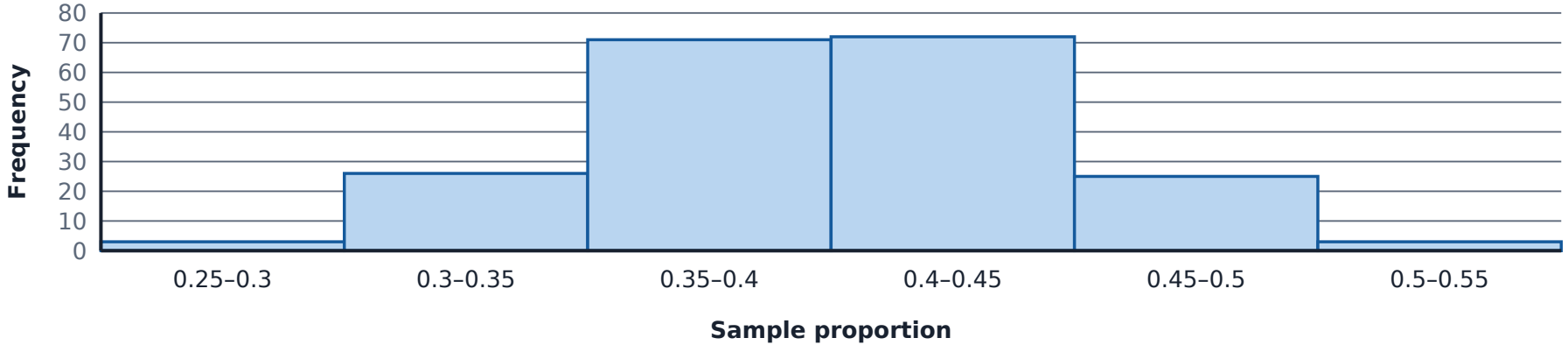
To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

The histogram shows 200 simulated sample proportions. How many of them are between 0.35 and 0.45 (bins 0.35-0.40 and 0.40-0.45)?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

A poll of a random sample reports that 54% of voters support a measure, with a margin of error of 3 percentage points. Which conclusion is best supported?

A. Exactly 54% of all voters support the measure.

B. Support could plausibly be anywhere from 0% to 100%.

C. The poll shows that support is below 50%.

D. A majority plausibly supports it: 51% to 57% are all above 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 1

Compute a sample proportion and name it • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 250 students, 90 walk to school. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

True or false?

A sample proportion is a statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

The proportion of all voters in a state who favor a law is a

A. parameter

B. statistic

C. margin of error

D. sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5

In a random sample of 80 phones, 6 were defective. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Different random samples always give the same sample proportion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 2

Read a standard deviation as typical variation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

True or false?

A standard deviation measures how far values typically are from their mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

(Hypothetical simulation results.) Simulated proportions from samples of size 100 have SD 0.05; from samples of size 400, SD 0.025. Which vary less?

A. cannot tell

B. the proportions from samples of size 100

C. the proportions from samples of size 400

D. they vary the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

Compute 2×0.031.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

True or false?

A larger standard deviation means the values are more tightly clustered.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

About what percent of values in a roughly normal distribution lie within 2 SDs of the mean?

A. 68%

B. 95%

C. 50%

D. 99.7%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 1

See how sample size changes the margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

For a sample proportion near 0.5, the margin of error is about $2 \cdot \sqrt{(0.5 \cdot 0.5/n)}$. Find it for $n = 400$ and $n = 100$.

$n = 400$: _____

$n = 100$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

True or false?

Making the sample 4 times as large cuts the margin of error about in half.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

A pollster wants a smaller margin of error. What should change?

A. take a larger random sample

B. ask only volunteers

C. round the results

D. take a smaller random sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5

(Hypothetical simulation results.) Simulated proportions from samples of size 900 have SD 0.016. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

True or false?

A larger sample removes the bias of a poorly worded question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 2

Read margins of error in poll reports • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

Candidate A has 48% and candidate B has 45%, each with a margin of error of 4 percentage points. What is the best conclusion?

A. Both candidates have under 50%, so both lose.

B. No clear leader: the two intervals overlap a lot.

C. B is certainly behind A by at least 3 points.

D. A is certainly ahead of B by exactly 3 points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5

A report gives 62% with a margin of error of 5 percentage points. Give the interval in percents.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

True or false?

A margin of error describes only random sampling variation, not bias from how the sample was chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

A website poll of 10,000 visitors reports 70% ± 1%. Why is the margin of error misleading?

A. The sample is too small for a margin of error.

B. A margin of error must be at least 5%.

C. Percents cannot have a margin of error.

D. The visitors chose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5

True or false?

If two candidates' intervals overlap, the poll proves that they are tied.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

Estimate a population mean with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Use a margin of error	**HSS-IC.B.4**
BACKWARD 2	Compute a mean	**6.SP.B.5.c**
FORWARD 1	Compare variability of sample means	**HSS-IC.B.4**
FORWARD 2	Compare two treatments' means	**HSS-IC.B.5**

SET 28 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 8 students reported these hours of sleep: 7, 6.5, 8, 7.5, 6, 9, 7, 8. What is the sample mean?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

(Hypothetical simulation results.) Simulated random samples of 8 students gave sample means with standard deviation 0.15 hour. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

A sample mean is 7.4 hours with a margin of error of 0.3 hour. Give the interval of plausible values for the population mean.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

A study reports a mean of 7.4 hours of sleep with a margin of error of 0.3 hour, from a random sample. Which interpretation is correct?

A. Population means from 7.1 to 7.7 hours are plausible.

B. The population mean is exactly 7.4 hours.

C. Half of all students sleep 7.4 hours.

D. Every student sleeps between 7.1 and 7.7 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5

(Hypothetical simulation results.) In 1,000 simulated samples, the middle 95% of sample means ran from 41.2 to 44.8. Give the center of this interval and the margin of error (half its width).

Center: _____

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 1

Use a margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

An estimate is 0.40 with a margin of error of 0.06. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Margin of error = 2 × 0.05 = 0.1 when the simulated values have SD 0.05.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

Which estimate has the widest interval?

A. 20 ± 1

B. 20 ± 4

C. 20 ± 2

D. 20 ± 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5

An interval runs from 12 to 18. What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

True or false?

An interval from 0.30 to 0.40 has center 0.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 2

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 4, 6, 8, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5

Find the mean of 6.5, 7, 8.5, rounded to three decimal places.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5

True or false?

The mean of 2, 4, 9 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5

What is the mean of 10, 12, 14, 20?

A. 14

B. 56

C. 13

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

The mean of a data set must be one of its values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 1

Compare variability of sample means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Sample means from samples of size 9 have SD 2; from samples of size 36, SD 1. Which statement is correct?

A. Means of larger samples vary more, so the margin is larger.

B. Both sample sizes give the same margin of error.

C. Means of larger samples vary less, so the margin of error is smaller.

D. Sample size has no effect at all on sample means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

(Hypothetical simulation results.) Simulated sample means from samples of size 36 have SD 1. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

True or false?

Individual values vary more than means of samples of those values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

An estimate of 52 has a margin of error of 2. Is 55 a plausible population mean? Enter yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

True or false?

A margin of error corrects the bias of a convenience sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 2

Compare two treatments' means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plant food A: mean growth 13 cm; plant food B: mean growth 10 cm. What is the difference (A minus B)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

An experiment's difference in means could be due to the treatments or to chance. What tool checks chance?

A. a re-randomization simulation

B. a census

C. a larger margin of error

D. a scatter plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5

True or false?

A difference in means from a randomized experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

(Hypothetical simulation results.) In 200 re-randomizations, a difference of 3 or more occurred 4 times. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

True or false?

A difference that chance alone produces in only 2% of re-randomizations is significant under a 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

Compare two treatments with a randomization simulation

MAIN	Probability & Statistics main question	**HSS-IC.B.5**
BACKWARD 1	Decide whether a model is consistent with data	**HSS-IC.A.2**
BACKWARD 2	Explain random assignment	**HSS-IC.B.3**
BACKWARD 3	Compute means of small groups	**6.SP.B.5.c**
FORWARD 1	Use differences in either direction	**HSS-IC.B.5**
FORWARD 2	Evaluate a report of an experiment	**HSS-IC.B.6**

SET 29 • HSS-IC.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Ten plants were randomly assigned to plant food A or B. Find the mean growth for each group and the difference (A minus B).

Plant food	Growth of the 5 plants (cm)
A	14, 11, 13, 12, 15
B	10, 12, 9, 11, 8

Mean A: _____

Mean B: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

(Hypothetical simulation results.) In an experiment, 10 plants were randomly assigned to two plant foods (5 each), and the difference in mean growth was 3 cm. The 10 results were re-randomized into two groups of 5 a total of 200 times. A difference of 3 or more occurred 4 times. What proportion of the re-randomizations is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

(Hypothetical simulation results.) Ten plants were randomly assigned to plant food A or B (5 each); mean growth A minus B was 3 cm. In 200 re-randomizations, a difference (A minus B) of 3 or more occurred 4 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 3 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. The difference is significant: any difference is significant.

B. The difference is not significant: it happened 4 times by chance.

C. The difference is significant: chance gave it only 2% of the time.

D. The difference is not significant: the groups had only 5 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5

In an experiment, 10 plants were randomly assigned to plant food A or B (5 each). What does one re-randomization of the 10 growth results do?

A. Repeats the whole experiment with many more plants.

B. Swaps the means of the two original groups of plants.

C. Reshuffles the 10 results into groups of 5, ignoring food.

D. Grows 10 new plants, again using the two plant foods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

(Hypothetical simulation results.) In another plant-food experiment with random assignment, mean growth A minus B was 1.2 cm. In 200 re-randomizations, differences (A minus B) of 1.2 or more occurred 62 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 1.2 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. Not significant: re-randomizations can never be significant.

B. Significant: 62 is a large number of re-randomizations.

C. Not significant: chance gave such a difference 31% of the time.

D. Significant: the difference 1.2 is greater than 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 1

Decide whether a model is consistent with data • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

True or false?

A result that would happen rarely under a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

(Hypothetical simulation results.) A result at least as extreme happened in 3 of 150 simulated trials. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

Under the 5% rule, a result as extreme as the data occurred in 12% of trials. The decision is

A. the model is proven

B. consistent with the model

C. not consistent with the model

D. no decision

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5

True or false?

Under the 5% rule, 4% of trials means the model is consistent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

Which proportion of simulated results gives the strongest evidence against a model?

A. 0.001

B. 0.2

C. 0.04

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 2

Explain random assignment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Random assignment tends to make the treatment groups similar before treatment, although chance differences remain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

In an experiment, plants were assigned to two plant foods at random. Why?

A. so that more of the plants grow well

B. so that no plant needs to be measured

C. so growth differences can be credited to food

D. so the two group means come out equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

True or false?

In an experiment, subjects choose which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

Ten plants are numbered 1-10. Which assignment is random?

A. put the 5 tallest plants on food A

B. let the gardener choose

C. draw 5 numbers from a hat for food A

D. put plants 1-5 on food A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5

True or false?

Random assignment is what allows an experiment to support a conclusion about cause.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 3

Compute means of small groups • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 14, 11, 13, 12, 15.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5

Find the mean of 10, 12, 9, 11, 8.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5

True or false?

The difference of the means 13 and 10 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5

Find the mean of 2.5, 3, 3.5, 5.

A. 3

B. 3.5

C. 3.25

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5

True or false?

The mean of 8, 9, 10, 11, 12 is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 1

Use differences in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

(Hypothetical simulation results.) In 200 re-randomizations, differences of 3 or more occurred 4 times and differences of −3 or less occurred 5 times. What proportion is at least 3 away from 0 in either direction? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

True or false?

Under a 5% rule, a proportion of 0.045 means the difference is significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

With more plants per group, re-randomized differences tend to be

A. always positive

B. closer to 0

C. farther from 0

D. exactly 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5

Treatment mean 31, control mean 28. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

True or false?

A significant difference proves that the treatment helps every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 2

Evaluate a report of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company reports: "Our drink improved scores in a study." The study gave the drink to 10 volunteers and compared them with 10 who chose not to drink it. What is the main flaw?

A. Drinks cannot be studied in an experiment.

B. Subjects chose their own group: no random assignment.

C. Twenty subjects is too many for a study.

D. Scores of two groups cannot be compared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5

True or false?

A report should say whether a difference could plausibly be due to chance.

A. True

B. False

(Hypothetical simulation results.) A report says a new method raised mean scores from 70 to 71, "a significant improvement." Re-randomization shows such differences 40% of the time. What is the best response?

A. The new method actually lowered scores.

B. Proven true: the study was an experiment.

C. Supported: 71 is higher than 70.

D. Not supported: chance often gives such differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5

True or false?

A randomized experiment on adults automatically applies to children.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

Which detail most strengthens a report's causal claim?

A. subjects were randomly assigned to treatments

B. the study had a long and technical name

C. the researchers felt confident in the result

D. the report used colorful, eye-catching graphs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5

Evaluate the conclusions of reports based on data

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Know what each design can support	**HSS-IC.B.3**
BACKWARD 2	Recognize lurking variables	**HSS-ID.C.9**
FORWARD 1	Identify the type of bias in a report	**HSS-IC.B.6**
FORWARD 2	Weigh several flaws and decide what a report shows	**HSS-IC.B.6**

SET 30 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says: "Teens who eat breakfast get better grades, so eating breakfast raises grades." The data came from a survey of 2,000 randomly chosen teens. What is the main problem?

A. A survey shows association, not that breakfast causes better grades.

B. A random sample of 2,000 teens is far too small for any survey.

C. Grades cannot be compared between different groups of teens.

D. Random samples cannot be used to describe a population of teens.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5

A magazine asked readers to mail in whether they liked their jobs; 80% of the 3,000 replies said no. The magazine reports that "80% of workers dislike their jobs." What is the main problem?

A. Readers chose to reply, so the sample is likely biased.

B. Three thousand replies is too few to report.

C. The percent 80% was computed incorrectly.

D. Job satisfaction cannot be surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5

A survey asked, "Don't you agree that the unfair new tax should be repealed?" What is the problem?

A. The question is too short to be answered.

B. The wording pushes people toward one answer.

C. The sample must have been too large.

D. Tax questions cannot appear in surveys.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5

A randomized experiment found that a reading program raised scores of 3rd graders in one district. Which conclusion is best supported?

A. The program will raise scores for students of every age everywhere.

B. The program caused higher scores for students like those in the study.

C. The program is associated with scores but did not cause any change.

D. Nothing can be concluded, because only one district was studied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5

A report on 12 patients says a new treatment "works for everyone," because 9 of the 12 improved. Which criticism is most important?

A. Twelve patients is too many to study at once.

B. The fraction 9/12 should have been written 12/9.

C. No comparison group, and 3 of the 12 did not improve.

D. Improvement cannot be measured in patients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 1

Know what each design can support • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Of a sample survey, an observational study and a randomized experiment, the randomized experiment is the design built to support a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

Which design lets results be generalized to a population?

A. a group of volunteers

B. a convenience sample

C. an experiment on friends

D. a random sample from that population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5

True or false?

An observational study can prove cause if it is large enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

A study used volunteers and random assignment. Its conclusions can include

A. nothing at all

B. cause, for the whole population

C. cause, for subjects like the volunteers

D. association only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5

True or false?

Random sampling and random assignment serve different purposes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 2

Recognize lurking variables • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more police officers have more crimes. What is a likely lurking variable?

A. the number of officers

B. crime reports

C. city population

D. police uniforms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

True or false?

A lurking variable affects both variables in an association.

A. True

B. False

Students who take music lessons score higher in math. Which is a possible lurking variable?

A. family income

B. the music teacher's name

C. the number of notes played

D. math test length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

True or false?

If an association is strong, there can be no lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

True or false?

Random assignment protects an experiment from lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 1

Identify the type of bias in a report • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A phone survey calls only landlines during work hours. Which bias is most likely?

A. no bias, because calls are made at random

B. response bias from the wording of questions

C. voluntary response bias from self-selection

D. undercoverage of people away or with only cell phones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5

Many people chosen for a survey refuse to answer. Which bias is this?

A. nonresponse bias

B. measurement error

C. wording bias

D. voluntary response bias

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5

True or false?

A larger sample removes bias caused by a poor sampling method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

A survey of mall shoppers is used to describe all residents' shopping habits. Which bias is this?

A. random sampling error only, with no bias

B. a convenience sample that is not representative

C. wording bias from a leading question

D. nonresponse bias from refusals to answer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5

True or false?

People may give socially acceptable answers instead of true ones when asked sensitive questions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 2

Weigh several flaws and decide what a report shows • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An app maker reports: "Users who used our app for 30 days lost 8 pounds on average." The users were volunteers and there was no comparison group. Which conclusion is justified?

A. Only that these volunteers lost weight; the app's role is unclear.

B. The app causes weight loss in everyone who uses it.

C. Every single user of the app lost exactly 8 pounds.

D. The study has shown that the app has no effect on weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5

True or false?

A report that gives no information about how its sample was chosen should be treated with caution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

Which change would most improve the app study?

A. ask the users whether they liked the app

B. randomly assign volunteers to use the app or not

C. extend the study from 30 days to 31 days

D. report the weight loss as a percent instead

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5

True or false?

A study funded by a company is always wrong about that company's product.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

Two reports disagree. Report 1 is a randomized experiment with 400 subjects; report 2 is a voluntary online poll of 5,000. Which deserves more weight for a causal claim?

A. equal weight

B. report 1

C. report 2

D. neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSS-IC.B.6 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Evaluate the statistics and displays in reports

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Use margins of error to compare results	**HSS-IC.B.4**
BACKWARD 2	Compute percent change	**7.RP.A.3**
FORWARD 1	Spot misleading features of displays	**HSS-IC.B.6**
FORWARD 2	Reason about conditional rates in reports	**HSS-CP.A.5**

SET 31 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A bar graph's vertical axis starts at 50. One bar shows 52 and another shows 55. How tall is the second bar compared with the first (ratio of drawn heights), and what is the ratio of the actual values? Give both ratios to two decimal places.

Ratio of drawn heights: _____

Ratio of values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

An ad says a risk "increased 80%," from 1 in 10,000 to 1.8 in 10,000. Which statement is correct?

A. The absolute risk rose by only 0.8 in 10,000.

B. Most people will now be affected by it.

C. The risk is now 80 out of every 100 people.

D. The risk went from 1% of people to 80%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5

A poll shows candidate A at 51% and B at 49%, with a margin of error of 3 percentage points. A headline says "Poll shows A has a reliable lead among all voters!" What is the best evaluation?

A. Misleading: the gap is well within the margin of error.

B. Correct: A has a majority of all the voters.

C. Misleading: B is certainly ahead of A.

D. Correct: 51% is more than 49%, so A leads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5

A report gives the mean income in a town as $95,000, but most households earn under $50,000 and a few earn millions. What is the best evaluation?

A. The report must have added incomes incorrectly.

B. The mean is the best summary for skewed data.

C. The range is the best measure of center.

D. The median would describe a typical household better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

A pictograph shows sales doubling by drawing a square icon twice as tall and twice as wide. How many times as large is the new icon's area?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 1

Use margins of error to compare results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

An estimate is 51% with a margin of error of 3 points. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

True or false?

If two intervals overlap a lot, a cautious reading is not to claim from them which value is larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

Which pair of estimates (each ± 2 points) clearly differs?

A. 48% and 50%

B. 40% and 50%

C. 49% and 51%

D. 45% and 47%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A margin of error of 3 points means the estimate is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

An interval is 46% to 52%. What is the estimate at its center?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 2

Compute percent change • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A value rises from 1 to 1.8. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5

A price rises from $40 to $50. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5

True or false?

A rise from 2 to 3 is a 50% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5

A count falls from 200 to 150. What is the percent decrease?

A. 33%

B. 75%

C. 50%

D. 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5

True or false?

A rise from 50 to 55 is a 5% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 1

Spot misleading features of displays • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A line graph of monthly sales has a vertical axis from 980 to 1,000. What effect does this have?

A. Large changes look very small.

B. It hides the months.

C. Small changes look very large.

D. It has no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5

True or false?

A pie chart whose percents add to 130% is misleading or in error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

A graph's vertical axis starts at 90. Bars show 92 and 98. What is the ratio of the drawn heights?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

True or false?

Using a 3-D effect on a pie chart never changes how large slices appear.

A. True

B. False

Which display choice is most honest for comparing two counts of 52 and 55?

A. a bar graph with the axis starting at 51

B. a 3-D pie chart

C. a bar graph with the axis starting at 0

D. icons of different widths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 2

Reason about conditional rates in reports • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

A report says "most car accidents happen within 5 miles of home, so driving near home is dangerous." What is the main flaw?

A. Car accidents cannot be counted accurately by anyone.

B. A distance of five miles is too short to be studied.

C. Cars are always safer when driven far from home.

D. It uses counts without the amount of driving near and far from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

True or false?

Knowing that 90% of people in a hospital are sick does not tell you the chance that a sick person is in the hospital.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

A test catches 95% of people who have a rare disease. A report concludes that 95% of people who test positive have it. What is the error?

A. Rare diseases cannot be tested for at all.

B. It confuses P(positive given disease) with P(disease given positive).

C. A test must catch 100% of cases before it is reported.

D. The 95% should have been written as a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5

True or false?

A higher number of accidents involving red cars proves that red cars are more dangerous.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

To compare the safety of two car colors fairly, which statistic is needed?

A. the number of different car colors sold

B. the average price of the cars of each color

C. accidents per car (or per mile) for each color

D. the total number of accidents for each color

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5

Describe sample spaces and events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	List a sample space with a list, table or tree	**7.SP.C.8.b**
BACKWARD 2	Find probabilities in a uniform model	**7.SP.C.7.a**
FORWARD 1	Describe events in multi-stage sample spaces	**HSS-CP.A.1**
FORWARD 2	Combine events with "and", "or" and "not"	**HSS-CP.A.1**

SET 32 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair coin is flipped and a fair six-sided die is rolled, independently. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

Two fair coins are flipped independently; the sample space is HH, HT, TH, TT. How many outcomes are in the event "at least one head"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

A fair coin is flipped twice, independently, and the sequence of results is recorded. Which is the sample space?

A. {H, T}

B. {HH, HT, TT}

C. {HH, TT}

D. {HH, HT, TH, TT}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5

Two fair dice are rolled independently. How many of the 36 equally likely outcomes are in the event "the sum is 7"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

A spinner has 8 equal sections numbered 1 to 8. Event A is "an even number greater than 3". How many outcomes are in A, and what is P(A)? Give a fraction in lowest terms.

Outcomes in A: _____

P(A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 1

List a sample space with a list, table or tree • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A bag has a red and a blue marble; a fair coin is flipped. How many outcomes (color, coin) are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5

True or false?

A tree diagram for flipping a coin 3 times (heads or tails each time) has 8 final branches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5

A sandwich shop offers 3 breads and 4 fillings. How many bread-filling choices are there?

A. 12

B. 81

C. 7

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The sample space for rolling two six-sided dice (as ordered pairs) has 12 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5

A spinner with 3 equal colors is spun twice. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair die is rolled. What is P(a number greater than 4)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5

A letter is chosen at random from the word MATH. What is P(vowel)?

A. 1/2

B. 1/3

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5

A number is chosen at random from 1 to 20. What is P(multiple of 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5

True or false?

A spinner with sections of different sizes is a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 1

Describe events in multi-stage sample spaces • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Three fair coins are flipped independently. How many outcomes are in the sample space, and how many have exactly 2 heads?

Outcomes: _____

Exactly 2 heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

Three fair coins are flipped independently. What is P(exactly 2 heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

True or false?

For two fair dice rolled independently, the event "doubles" contains 6 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

For two fair dice rolled independently, which event contains exactly 4 outcomes?

A. the sum is 7

B. the sum is 12

C. doubles

D. the sum is 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

Two fair dice are rolled independently. How many outcomes have a sum of 10 or more?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 2

Combine events with "and", "or" and "not" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A and B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A or B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

True or false?

The event "not A" contains every outcome of the sample space that is not in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

A fair six-sided die is rolled. A = "even". Which set is "not A"?

A. {2, 4, 6}

B. {1, 2, 3}

C. { }

D. {1, 3, 5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q4/5

True or false?

The event "A and B" always contains at least as many outcomes as "A or B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

Unions, intersections and complements of events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	Count the outcomes in an event	**HSS-CP.A.1**
BACKWARD 2	Find a probability by counting	**7.SP.C.8.a**
FORWARD 1	Use a Venn diagram of counts	**HSS-CP.A.1**
FORWARD 2	Apply the Addition Rule	**HSS-CP.B.7**

SET 33 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A and B, and what is P(A and B)? Give a fraction in lowest terms.

Outcomes: _____

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A or B, and what is P(A or B)? Give a fraction in lowest terms.

Outcomes: _____

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

A fair die is rolled. A = "even". What is P(not A)? Give a fraction in lowest terms.

P(not A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

A fair die is rolled. A = "even" and B = "greater than 3". Which describes the event {1, 3}?

A. A or B

B. A and B

C. not (A or B)

D. not A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

A card is drawn at random from a standard 52-card deck. H = "heart", F = "face card (jack, queen or king)". Find P(H and F) and P(H or F). Give a fraction in lowest terms.

P(H and F): _____

P(H or F): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 1

Count the outcomes in an event • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A number from 1 to 10 is chosen. How many outcomes are in "multiple of 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

True or false?

For one roll of a fair six-sided die, the event "less than 3" is {1, 2}.

A. True

B. False

A letter is chosen from the word PROBABILITY. How many outcomes (letter positions) are in "the letter B"?

A. 11

B. 1

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5

How many cards in a standard 52-card deck are face cards?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

A standard deck has 4 aces and 13 hearts, so it has 17 cards that are aces or hearts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 2

Find a probability by counting • Grade 7 • Statistics & Probability • 7.SP.C.8.a

Two fair coins are flipped independently. What is P(two heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5

A fair six-sided die is rolled. What is P(1 or 2)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

The probability of an event plus the probability of its complement is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(king)?

A. 1/4

B. 1/52

C. 4/13

D. 1/13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5

True or false?

If P(A) = 0.3, then P(not A) = 0.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 1

Use a Venn diagram of counts • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. How many play soccer only, and how many play neither?

Soccer only: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. A student is chosen at random. What is P(soccer or basketball)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

True or false?

In a Venn diagram, the overlap of two circles shows "A and B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. What is P(not soccer)?

A. 3/5

B. 2/5

C. 7/30

D. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5

True or false?

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. Adding 18 and 12 gives the number who play soccer or basketball.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 2

Apply the Addition Rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

A card is drawn at random from a standard 52-card deck. Find P(king or heart). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

If A and B cannot happen together, P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

P(A) = 0.6, P(B) = 0.3, P(A or B) = 0.7. What is P(A and B)?

A. 0.9

B. 0.2

C. 0.1

D. 0.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5

True or false?

P(A or B) can be greater than 1 if A and B are both likely.

A. True

B. False

Decide independence with the product rule

MAIN	Probability & Statistics main question	**HSS-CP.A.2**
BACKWARD 1	Multiply probabilities	**5.NF.B.4**
BACKWARD 2	Find P(A and B) by counting	**HSS-CP.A.1**
FORWARD 1	Use independence for several events	**HSS-CP.A.2**
FORWARD 2	Interpret independence with conditional probability	**HSS-CP.A.3**

SET 34 • HSS-CP.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

$P(A) = 0.4$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.2$. Find $P(A)\cdot P(B)$, and decide whether A and B are independent (enter yes or no).

$P(A)\cdot P(B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.25. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

A and B are independent with P(A) = 0.7 and P(B) = 0.2. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

A fair six-sided die is rolled. A = "even" and B = "1 or 2". Are A and B independent?

A. No: A and B share the outcome 2.

B. No: P(A and B) = 1/6, not 1/2.

C. Yes: P(A and B) = 1/6 = (1/2)(1/3).

D. Yes: A and B have no outcomes in common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5

In the survey table, a person is chosen at random. M = "man" and Y = "said yes". Find P(M and Y), and decide whether M and Y are independent (enter yes or no).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(M and Y): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 1

Multiply probabilities • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

Compute (1/2)(1/3). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5

Compute 0.4 · 0.5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5

True or false?

$(2/3)(3/4) = 1/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5

Compute 0.3 · 0.6.

A. 1.8

B. 0.36

C. 0.18

D. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5

True or false?

$0.7 \cdot 0.2 = 1.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 2

Find P(A and B) by counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled. What is P(even and greater than 3)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

A fair coin is flipped and a fair die is rolled, independently. What is P(heads and 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

True or false?

For one roll of a fair six-sided die, P(odd and even) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

Two fair coins are flipped independently. What is P(first is heads and second is tails)?

A. 3/4

B. 1/4

C. 1

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5

True or false?

For one roll of a fair six-sided die, P(less than 3 and odd) = 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 1

Use independence for several events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A fair coin is flipped 3 times, independently. What is P(all three heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

Each of 3 independent lights works with probability 0.9. What is P(all 3 work), and P(at least one fails)?

All work: _____

At least one fails: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

True or false?

For independent events, P(A and B and C) = P(A)·P(B)·P(C).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

A fair die is rolled twice, independently. What is P(two sixes)?

A. 1/6

B. 1/36

C. 2/6

D. 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

True or false?

If P(A and B) = P(A)·P(B), then A and B cannot happen together.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 2

Interpret independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.2 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

True or false?

If A and B are independent, P(A given B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

P(A) = 0.4 and P(A given B) = 0.4. What can be concluded?

A. A and B are independent.

B. P(B) = 0.4.

C. B always happens.

D. A and B are mutually exclusive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.18. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

If P(A given B) is greater than P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

Compute conditional probability as P(A and B)/P(B)

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Divide decimals and fractions	**6.NS.B.3**
BACKWARD 2	Count outcomes in a restricted sample space	**HSS-CP.A.1**
FORWARD 1	Compute conditional probability in both directions	**HSS-CP.A.3**
FORWARD 2	Test independence with conditional probability	**HSS-CP.A.3**

SET 35 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

A fair die is rolled. What is P(6 given the roll is even)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

Two fair dice are rolled independently. What is P(doubles given the sum is 8)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

On a given day, P(rain) = 0.3 and P(rain and wind) = 0.12. What is P(wind given rain)?

A. 0.4

B. 2.5

C. 0.12

D. 0.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

P(B) = 0.25 and P(A given B) = 0.6. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 1

Divide decimals and fractions • Grade 6 • The Number System • 6.NS.B.3

Compute 0.12 ÷ 0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5

Compute 0.18 ÷ 0.6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5

True or false?

$0.2 \div 0.5 = 0.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5

Compute $0.05 \div 0.2$.

A. 0.25

B. 0.01

C. 2.5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

$0.3 \div 0.6 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 2

Count outcomes in a restricted sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled and shows an even number. How many outcomes remain possible?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

Two fair dice are rolled independently and the sum is 8. How many outcomes give a sum of 8?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a heart leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

A fair six-sided die is rolled and shows a number greater than 2. How many of the remaining outcomes are odd?

A. 2

B. 3

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a face card leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSS-CP.A.3

FORWARD 1

Compute conditional probability in both directions • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

A card is drawn at random from a standard 52-card deck. Find P(king given face card) and P(face card given king). Give a fraction in lowest terms.

P(king given face card): _____

P(face card given king): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

True or false?

P(A given B) and P(B given A) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

P(A) = 0.5, P(B) = 0.2 and P(A and B) = 0.1. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

Which formula gives P(B given A)?

A. P(A)/P(B)

B. P(A)·P(B)

C. P(A and B)/P(A)

D. P(A and B)/P(B)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5

True or false?

P(A given B) = P(B given A) exactly when P(A) = P(B), provided P(A and B) > 0.

A. True

B. False

SET 35 • HSS-CP.A.3

FORWARD 2

Test independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A) = 0.6, P(B) = 0.5, P(A and B) = 0.3. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

True or false?

If P(A given B) = P(A), then also P(B given A) = P(B), provided P(A) > 0.

A. True

B. False

Two cards are drawn at random, without replacement, from a standard 52-card deck. Are "first card is an ace" and "second card is an ace" independent?

A. No: two aces can never be drawn one after another.

B. Yes: the deck always contains exactly 4 aces.

C. Yes: each card in the deck is equally likely to be drawn.

D. No: the first draw changes the chance for the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5

P(A) = 0.4, P(B) = 0.5, P(A and B) = 0.1. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

True or false?

If P(A given B) = 0.2 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

Interpret independence through conditional probability

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Compute a conditional probability	**HSS-CP.A.3**
BACKWARD 2	Test independence with the product rule	**HSS-CP.A.2**
FORWARD 1	Check both directions of independence	**HSS-CP.A.3**
FORWARD 2	Approximate conditional probabilities from data	**HSS-CP.A.4**

SET 36 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.2$. Find $P(A \text{ given } B)$, and decide whether A and B are independent (enter yes or no).

$P(A \text{ given } B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

$P(A) = 0.3$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.1$. Find $P(A \text{ given } B)$, and decide whether A and B are independent (enter yes or no).

$P(A \text{ given } B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

A and B are independent. Which statement is correct?

A. Whenever B happens, then A must also happen.

B. P(A) and P(B) must be equal to each other.

C. A and B can never happen at the same time as each other.

D. Knowing that B happened does not change the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5

Two cards are drawn at random from a standard 52-card deck. Find P(second is an ace given the first is an ace) when the first card is replaced, and when it is not replaced. Give a fraction in lowest terms.

With replacement: _____

Without replacement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

Which pair of events is independent?

A. a student studies; the student passes the test

B. first card drawn is an ace; second card (not replaced) is an ace

C. it rains today; the ground is wet today

D. a die shows 6; a separate coin shows heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 1

Compute a conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.06 and P(B) = 0.3. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

P(A and B) = 0.25 and P(A) = 0.5. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

True or false?

P(A given B) = P(A and B)/P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

P(A and B) = 0.1, P(B) = 0.4. Find P(A given B).

A. 0.4

B. 0.25

C. 4

D. 0.04

True or false?

P(A given B) = P(A and B)·P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 2

Test independence with the product rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

True or false?

If P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2, A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

P(A) = 0.2 and P(B) = 0.5. If A and B are independent, what is P(A and B)?

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

$P(A) = 0.3$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$. Are A and B independent?

A. no, because $0.2 > 0$

B. yes, because $0.3 < 0.4$

C. yes, because $0.2 < 1$

D. no, because $0.3 \cdot 0.4 = 0.12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5

True or false?

Mutually exclusive events with positive probabilities are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

P(A) = 0.6 and P(B) = 0.5 for independent events. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 1

Check both directions of independence • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A) = 0.6, P(B) = 0.25 and P(A and B) = 0.15. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

True or false?

If P(A given B) = P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

P(A) = 0.5, P(A given B) = 0.5, and P(B) = 0.3. What is P(B given A)?

A. 0.15

B. 0.6

C. 0.5

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5

A and B are independent, P(A) = 0.7 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

True or false?

If P(B given A) = 0.25 and P(B) = 0.5, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 2

Approximate conditional probabilities from data • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the survey table, a person is chosen at random. Find P(yes given man) and P(yes).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(yes given man): _____

P(yes): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

True or false?

In the survey table, gender and answer are independent.

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

In the driving table, find P(drives given junior) and P(drives).

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P(drives given junior): _____

P(drives): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

True or false?

In the driving table, grade and driving are independent.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

In the driving table, which is P(junior given drives)?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. 1/2 **B.** 3/10 **C.** 2/3 **D.** 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

Construct two-way frequency tables

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Fill in a two-way frequency table	**8.SP.A.4**
BACKWARD 2	Read joint and marginal frequencies	**HSS-ID.B.5**
FORWARD 1	Check statements against a built table	**HSS-CP.A.4**
FORWARD 2	Use the table as a sample space	**HSS-CP.A.4**

SET 37 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

Of 200 students, 120 are juniors and 80 are seniors. 90 students drive to school, including 60 juniors. Complete the two-way table: juniors who do not drive, seniors who drive, seniors who do not drive.

Juniors, do not drive: _____

Seniors, drive: _____

Seniors, do not drive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5

Of 150 adults, 90 own a car, 40 own a bike, and 25 own both. In a two-way table (car yes/no by bike yes/no), how many own a car but no bike, and how many own neither?

Car, no bike: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

Of 200 students, 120 are juniors, and 60 of the juniors drive. What percent of juniors drive?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

80 patients took part in a study. Which description matches the table?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 50 patients got the placebo and the other 30 got the drug.

B. 50 patients improved, and 35 of those improved patients got the drug.

C. 50 got the drug (35 improved); 12 of the placebo patients improved.

D. 35 patients got the drug and 12 patients got a placebo instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5

80 patients: 50 got a drug and 30 a placebo; 35 drug patients and 12 placebo patients improved. How many drug patients and how many placebo patients did not improve?

Drug, did not: _____

Placebo, did not: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 1

Fill in a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

A table has row totals 40 and 60. One cell in the first row is 25. What is the other cell in that row?

Cell: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5

True or false?

In a two-way table, the column totals add to the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5

Of 50 people, 30 are adults. 12 of the adults wear glasses. How many adults do not?

A. 18

B. 12

C. 38

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5

A two-way table with two columns classifies 100 objects. One column total is 45. What is the other column total?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A cell can be found from its row total and the other cells in that row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 2

Read joint and marginal frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the drug table, how many patients improved in total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Improved: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

In the drug table, what fraction of all patients got the drug and improved? Give lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

True or false?

In the drug table, 30 patients got the placebo.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

In the drug table, which number is a marginal total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 12 **B.** 50 **C.** 35 **D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

True or false?

In the drug table, 35/47 is the fraction of drug patients who improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 1

Check statements against a built table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

True or false?

In the drug table, a greater percent of drug patients than placebo patients improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5

In the drug table, what percent of drug patients improved, and what percent of placebo patients improved?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Drug: _____

Placebo: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

A report says "most patients who improved got the drug." Using the drug table, is this correct?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. No: only 35 patients got the drug.

B. Yes: 70% of all patients improved.

C. Yes: 35 of the 47 who improved got the drug.

D. No: 50 patients got the drug.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5

Of 120 voters, 70 are under 40. 45 of all voters support a plan, including 30 of those under 40. How many voters 40 or older do not support it?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

True or false?

A two-way table can be built when each object is classified by only one category.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 2

Use the table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A patient is chosen at random from the drug table. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

A patient is chosen at random from the drug table. What is P(improved given drug)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

In the drug table, improving and treatment are independent.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

A patient is chosen at random from the drug table. What is P(drug given improved)?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 47/80 **B.** 35/47 **C.** 7/16 **D.** 35/50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5

A patient is chosen at random from the drug table. What is P(placebo and did not improve)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

Use two-way tables to find conditional probabilities and judge independence

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Compute conditional relative frequencies	**HSS-ID.B.5**
BACKWARD 2	Compare P(A given B) with P(A)	**HSS-CP.A.3**
FORWARD 1	Use larger two-way tables	**HSS-CP.A.4**
FORWARD 2	Explain table results in everyday language	**HSS-CP.A.5**

SET 38 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the driving table. Find P(drives given junior) and P(drives given senior). Give decimals.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Given junior: _____

Given senior: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5

In the driving table, are "drives" and "junior" independent?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. No: P(drives given junior) ≠ P(drives given senior).

B. Yes: every student is a junior or a senior.

C. Yes: 60 juniors drive and 60 do not.

D. No: more juniors than seniors were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5

A student is chosen at random from the sports table. Find P(plays a sport) and P(plays a sport given 9th grade). Give decimals.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

P(sport): _____

P(sport given 9th): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

In the sports table, are "plays a sport" and "9th grade" independent?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. Yes: P(sport given 9th) = P(sport) = 0.6.

B. No: 30 is not equal to 42.

C. No: the grades have different sizes.

D. Yes: more students play than do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5

A student who drives is chosen at random from the driving table. What is P(junior given drives)? Give a fraction in lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 1

Compute conditional relative frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the driving table, what fraction of seniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

True or false?

In the sports table, 42/70 of 10th graders play a sport.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

In the sports table, what fraction of students who play a sport are in 10th grade?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. 5/12 **B.** 7/12 **C.** 7/20 **D.** 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

In the driving table, what percent of all students drive?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A row conditional frequency divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 2

Compare P(A given B) with P(A) • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

If P(A given B) = P(A), A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

P(A) = 0.45 and P(A given B) = 0.5. Are A and B independent?

A. only if P(B) = 0.5

B. yes

C. no

D. cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5

P(A and B) = 0.3 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

True or false?

P(A given B) is always less than P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

P(A and B) = 0.18 and P(B) = 0.3. If P(A) = 0.6, are A and B independent?

A. yes

B. only if P(A) = 0.3

C. cannot tell

D. no

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 1

Use larger two-way tables • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the music table. Find P(rock) and P(rock given high school). Give decimals.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P(rock): _____

P(rock given high school): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

True or false?

In the music table, school level and favorite type of music are independent.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

In the music table, what is P(middle school given pop)? Give a fraction in lowest terms.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

In the music table, what is P(other given middle school)?

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. 1/5 **B.** 2/25 **C.** 2/5 **D.** 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

In the music table, P(pop given middle school) is greater than P(pop given high school).

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 2

Explain table results in everyday language • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

In the driving table, P(drives given junior) = 0.5 and P(drives given senior) = 0.375. Which sentence says this in everyday language?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Most of the students who drive to school are seniors.

B. Half of the juniors drive, but fewer than 4 in 10 seniors drive.

C. Juniors and seniors are equally likely to drive.

D. Half of all the students who drive are juniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5

True or false?

"Among students who drive, two-thirds are juniors" describes P(junior given drives).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

Which statement describes independence in the sports table?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. A student's grade does not change the chance of playing a sport.

B. Every student in the table plays exactly one sport.

C. The two grades have the same number of students.

D. Students in 10th grade play more sports than 9th graders.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

"Most juniors drive" and "most drivers are juniors" mean the same thing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

"Among seniors, 3 in 8 drive." Which probability is this?

A. P(senior given drives)

B. P(drives given senior)

C. P(senior and drives)

D. P(senior)

Conditional probability and independence in everyday language

MAIN	Probability & Statistics main question	**HSS-CP.A.5**
BACKWARD 1	Compute and read conditional probabilities	**HSS-CP.A.3**
BACKWARD 2	Read conditional rates in a table	**HSS-CP.A.4**
FORWARD 1	Translate statements and compute	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as a fraction of outcomes	**HSS-CP.B.6**

SET 39 • HSS-CP.A.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Which statement describes a conditional probability?

A. One in four cards in a deck is a heart.

B. Among smokers, the chance of lung disease is higher.

C. A fair coin lands heads half of the time.

D. The chance of rain tomorrow is 30 percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5

"Most people with lung disease were smokers." Which probability does this sentence say is high?

A. P(lung disease)

B. P(smoker given lung disease)

C. P(smoker and lung disease)

D. P(lung disease given smoker)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5

Which pair of events is most likely to be independent?

A. heavy rain today; a baseball game being canceled

B. a student's shirt color; the school bus being late

C. studying hard for a test; passing that test

D. a person's height; the person's shoe size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5

A fair coin has landed heads 5 times in a row. What is the probability of heads on the next flip?

A. 1/32

B. 1/2

C. less than 1/2

D. more than 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5

In a town, 40% of adults own a dog, and among dog owners 70% walk daily. What percent of all adults are dog owners who walk daily?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

SET 39 • HSS-CP.A.5

BACKWARD 1

Compute and read conditional probabilities • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.28 and P(A) = 0.4. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

True or false?

"The probability of B given A" means the chance of B when A is known to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

P(passing given studying) is the probability that

A. a student passes

B. a student passes, among students who study

C. a student studies, among students who pass

D. a student both studies and passes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5

P(B) = 0.5 and P(A given B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

P(rain given clouds) is the same as P(clouds given rain).

A. True

B. False

SET 39 • HSS-CP.A.5

BACKWARD 2

Read conditional rates in a table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the driving table, what fraction of juniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

True or false?

In the driving table, "among drivers, 2 in 3 are juniors" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

In the driving table, which sentence is correct?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Among seniors, 3 in 8 drive.

B. Among seniors, 1 in 2 drive.

C. Among juniors, 3 in 8 drive.

D. Among drivers, 3 in 8 are seniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5

In the driving table, what fraction of drivers are seniors? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

True or false?

In the driving table, "among seniors, most drive" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 1

Translate statements and compute • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

"If it is a school day, there is an 80% chance that Sam takes the bus." Which notation matches?

A. P(bus given school day) = 0.8

B. P(bus) = 0.8

C. P(school day given bus) = 0.8

D. P(bus and school day) = 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5

5 of 7 days are school days, and on a school day Sam takes the bus with probability 0.8. What is P(school day and bus)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

"Knowing it is a school day changes the chance that Sam takes the bus" means the events are not independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

A test is positive for 95% of people with a disease. Which sentence states this correctly?

A. 95% of all people have the disease.

B. 95% of all people test positive.

C. Among people with the disease, 95% test positive.

D. Among people who test positive, 95% have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5

True or false?

"Among people who test positive, 95% have the disease" follows from "95% of people with the disease test positive".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

A fair six-sided die is rolled. Given the roll is odd, what is P(the roll is 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

A card drawn at random from a standard 52-card deck is red. What is P(it is a king)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

True or false?

P(A given B) can be found as (outcomes in both A and B)/(outcomes in B) in a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

A number from 1 to 10 is even. What is P(it is greater than 6)?

A. 1/2

B. 2/5

C. 2/3

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5

True or false?

Given the roll of a fair six-sided die is even, P(roll is 3) = 1/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

Find conditional probability as a fraction of outcomes

MAIN	Probability & Statistics main question	**HSS-CP.B.6**
BACKWARD 1	Count outcomes in events	**HSS-CP.A.1**
BACKWARD 2	Use the definition of conditional probability	**HSS-CP.A.3**
FORWARD 1	Condition on "at least one"	**HSS-CP.B.6**
FORWARD 2	Multiply along a chain of conditions	**HSS-CP.B.8**

SET 40 • HSS-CP.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled independently. The first die shows 5. What is P(the sum is 10 or more)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

A card drawn at random from a standard 52-card deck is red. What is P(it is an ace)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

A patient from the drug table is chosen at random and got the placebo. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

A number from 1 to 10 is chosen at random and is odd. What is P(it is prime)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

From the drug table, P(improved given placebo) = 2/5. What does this mean in the model?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. Of the patients who improved, 2 in 5 got the placebo.

B. 2 in 5 of all patients got the placebo and improved.

C. Of the patients given the placebo, 2 in 5 improved.

D. 2 in 5 of all patients got the placebo.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 1

Count outcomes in events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

How many red cards are in a standard 52-card deck?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

Two fair dice rolled independently: how many outcomes have the first die equal to 5?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

True or false?

A standard 52-card deck has 2 red aces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

How many numbers from 1 to 10 are odd and prime?

A. 2

B. 5

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A standard 52-card deck has 4 red kings.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 2

Use the definition of conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 1/12 and P(B) = 1/4. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, P(A and B)/P(B) equals (count of A and B)/(count of B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

P(A and B) = 0.1 and P(B) = 0.5. Find P(A given B).

A. 5

B. 0.5

C. 0.05

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

P(A and B) = 2/36 and P(B) = 6/36. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

True or false?

P(A given B) = P(A and B)/P(A).

A. True

B. False

SET 40 • HSS-CP.B.6

FORWARD 1

Condition on "at least one" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled independently and at least one shows 6. What is P(the sum is 7)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

A family has two children; each is equally likely to be a girl or a boy. Given at least one is a girl, what is P(both are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two fair coins are flipped independently. Given at least one shows heads, P(both heads) = 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

Two fair dice are rolled independently and the sum is at least 11. What is P(doubles)?

A. 1/36

B. 1/6

C. 1/2

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5

True or false?

Conditioning on an event removes the outcomes outside it from the sample space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

FORWARD 2

Multiply along a chain of conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 4 red and 2 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

True or false?

P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

P(A) = 0.8 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(second is a heart given the first is a heart)?

A. 1/4

B. 13/52

C. 12/51

D. 13/51

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5

True or false?

For draws without replacement, P(second red given first red) equals P(first red).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

Apply the Addition Rule

MAIN	Probability & Statistics main question	**HSS-CP.B.7**
BACKWARD 1	Count a union without double counting	**HSS-CP.A.1**
BACKWARD 2	Add and subtract fractions	**5.NF.A.1**
FORWARD 1	Use the rule in reverse and with exclusive events	**HSS-CP.B.7**
FORWARD 2	Combine the Addition Rule with complements	**HSS-CP.B.7**

SET 41 • HSS-CP.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.45, P(B) = 0.35 and P(A and B) = 0.15. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

A card is drawn at random from a standard 52-card deck. Find P(red or king). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

Two fair dice are rolled independently. Find P(doubles or sum of 8). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

Of 200 students, 120 take Spanish, 70 take art, and 40 take both. A student is chosen at random. What is P(Spanish or art)?

A. 3/4

B. 7/10

C. 19/20

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5

For a randomly chosen shopper, P(buys bread or milk) = 0.65. What does this mean?

A. About 65% of shoppers buy bread, milk, or both.

B. About 65% of shoppers buy both bread and milk.

C. About 65% of shoppers buy neither one.

D. About 65% of shoppers buy exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 1

Count a union without double counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled. How many outcomes are "even or less than 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

True or false?

In a standard 52-card deck, the number of cards that are hearts or face cards is $13 + 12 - 3 = 22$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

In a standard 52-card deck, how many cards are aces or spades?

A. 4

B. 13

C. 16

D. 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5

How many numbers from 1 to 20 are multiples of 2 or multiples of 5?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The number of outcomes in "A or B" is always the count of A plus the count of B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 2

Add and subtract fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Compute 1/2 + 1/13 − 1/26. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5

Compute 1/6 + 5/36 − 1/36. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5

True or false?

1/4 + 1/3 = 7/12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5

Compute 3/5 + 7/20 − 1/5.

A. 17/20

B. 1

C. 3/4

D. 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 1

Use the rule in reverse and with exclusive events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.5, P(B) = 0.4 and P(A or B) = 0.7. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

True or false?

If A and B are mutually exclusive, P(A or B) = P(A) + P(B).

A. True

B. False

A fair six-sided die is rolled. Find P(1 or 6). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

P(A) = 0.3, P(B) = 0.5, and A and B are independent. What is P(A or B)?

A. 0.15

B. 0.5

C. 0.65

D. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5

True or false?

If P(A) = 0.6 and P(B) = 0.7, then A and B could be mutually exclusive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 2

Combine the Addition Rule with complements • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A or B) = 0.65. What is P(neither A nor B)?

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

Of 200 students, 120 take Spanish, 70 take art and 40 take both. What is P(neither)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

P(not (A or B)) = 1 − P(A or B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(neither red nor a king)?

A. 6/13

B. 1/2

C. 7/13

D. 11/26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q4/5

True or false?

P(neither A nor B) = 1 − P(A) − P(B) in every case.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

Apply the general Multiplication Rule (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.8**
BACKWARD 1	Rearrange the conditional probability formula	**HSS-CP.A.3**
BACKWARD 2	Multiply fractions	**5.NF.B.4**
BACKWARD 3	Multiply for independent events	**HSS-CP.A.2**
FORWARD 1	Combine orders and longer chains	**HSS-CP.B.8**
FORWARD 2	Count arrangements to find probabilities	**HSS-CP.B.9**

SET 42 • HSS-CP.B.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(first red and second blue)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both aces)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

P(A) = 0.6 and P(B given A) = 0.5. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

A committee has 3 boys and 2 girls. Two members are chosen at random, one after the other. What is P(both girls)?

A. 2/5

B. 1/10

C. 4/25

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 1

Rearrange the conditional probability formula • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

From P(B given A) = P(A and B)/P(A), it follows that P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

P(A) = 0.4 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

P(B) = 0.5 and P(A given B) = 0.6. What is P(A and B)?

A. 0.1

B. 1.1

C. 0.83

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5

P(A and B) = 0.12 and P(A) = 0.3. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

P(A and B) = P(A) + P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 2

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

Compute (5/8)(4/7). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5

Compute (4/52)(3/51). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5

True or false?

(2/5)(1/4) = 1/10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5

Compute (3/8)(2/7).

A. 3/28

B. 6/15

C. 3/56

D. 5/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

(1/2)(1/3) = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 3

Multiply for independent events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A fair die is rolled twice, independently. What is P(both 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

True or false?

With replacement, P(red then red) from 5 red and 3 blue is (5/8)(5/8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

A fair coin is flipped and a fair die is rolled, independently. P(tails and even)?

A. 2/3

B. 1/4

C. 1/12

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5

Independent events have P(A) = 0.5 and P(B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

Without replacement, draws are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 1

Combine orders and longer chains • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(one of each color)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

Three cards are drawn at random, without replacement, from a standard 52-card deck. What is P(all three are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

True or false?

P(A and B) = P(B)·P(A given B) gives the same answer as P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

A bag has 4 green and 6 yellow marbles. Two are drawn at random without replacement. What is P(both yellow)?

A. 9/25

B. 2/5

C. 3/5

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5

True or false?

P(first red and second blue) equals P(first blue and second red) for 5 red and 3 blue drawn without replacement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 2

Count arrangements to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 2 marbles be chosen from 8 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

How many ways can 2 red marbles be chosen from 5 red?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

P(both red) = 10/28 agrees with (5/8)(4/7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

How many ordered ways can 2 of 8 marbles be drawn one after the other?

A. 16

B. 28

C. 64

D. 56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5

True or false?

Counting with combinations always gives a different probability than the Multiplication Rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

Count ordered arrangements with permutations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count outcomes with a tree or the counting principle	**7.SP.C.8.b**
BACKWARD 2	Multiply several whole numbers	**5.NBT.B.5**
FORWARD 1	Count arrangements with conditions	**HSS-CP.B.9**
FORWARD 2	Count selections where order does not matter	**HSS-CP.B.9**

SET 43 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Five runners race. In how many ways can first, second and third place be awarded?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

In how many ways can 6 different books be arranged on a shelf?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

A 4-digit code uses the digits 0-9 with no digit repeated. How many codes are possible?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

What is the value of 8P2 (the number of ordered arrangements of 2 objects from 8)?

A. 56

B. 28

C. 16

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q4/5

A club of 10 members chooses a president, a vice president and a treasurer. How many ways are possible?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 1

Count outcomes with a tree or the counting principle • Grade 7 • Statistics & Probability • 7.SP.C.8.b

An outfit is a shirt (4 choices) and pants (3 choices). How many outfits?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5

True or false?

A tree for 3 stages with 2 choices each has 8 paths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5

A meal is a main (5 choices), a side (3) and a drink (2). How many meals?

A. 30

B. 60

C. 15

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5

A license code is a letter (26 choices) followed by a digit (10 choices). How many codes?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5

True or false?

With 3 shirts and 2 pants there are 5 outfits.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 2

Multiply several whole numbers • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

Compute $10 \cdot 9 \cdot 8$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5

Compute $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5

True or false?

$5 \cdot 4 \cdot 3 = 60$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5

Compute $7 \cdot 6 \cdot 5$.

A. 343

B. 180

C. 42

D. 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

$4 \cdot 3 \cdot 2 \cdot 1 = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 1

Count arrangements with conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

In how many ways can the letters of MATH be arranged if M must be first?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

In how many ways can 5 people stand in a line if Ana and Ben must be at the two ends?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

True or false?

$nPr = n!/(n - r)!$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

How many 3-letter arrangements (no repeats) can be made from the letters A, B, C, D, E if the first letter is a vowel?

A. 60

B. 12

C. 48

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

True or false?

The number of ways to arrange 4 people in a line is $4 + 3 + 2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 2

Count selections where order does not matter • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 of 5 students be chosen for a team (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

True or false?

Each unordered group of 3 people can be ordered in 6 ways.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

Which situation is counted with combinations rather than permutations?

A. choosing 3 pizza toppings

B. awarding gold, silver and bronze

C. making a 4-digit code

D. arranging books on a shelf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5

A permutation count is 120 for choosing 3 of 6 in order. How many unordered groups is that?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

True or false?

There are more combinations than permutations of 3 objects from 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

Count unordered selections with combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count ordered arrangements	**HSS-CP.B.9**
BACKWARD 2	Represent and count outcomes of compound events	**7.SP.C.8.b**
FORWARD 1	Count selections from separate groups	**HSS-CP.B.9**
FORWARD 2	Turn counts into probabilities	**HSS-CP.B.9**

SET 44 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 students be chosen from 5 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

A pizza shop has 8 toppings. How many 3-topping pizzas are possible (no repeated toppings)?

Pizzas: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

How many 4-person committees can be chosen from 10 people?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

What is the value of 6C2?

A. 15

B. 12

C. 36

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q4/5

Seven people each shake hands once with every other person. How many handshakes are there?

Handshakes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 1

Count ordered arrangements • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ordered ways can 3 of 5 students be lined up?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

True or false?

$5P3 = 60$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

How many ordered pairs of different people can be formed from 7 people?

A. 42

B. 21

C. 49

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-B1-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5

How many ways can 3 people be arranged in order?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

True or false?

$10P4 = 210$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 2

Represent and count outcomes of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

How many 2-letter groups (order does not matter) can be chosen from the letters A, B, C?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5

True or false?

From A, B, C, D, the groups AB and BA are the same group.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5

How many 2-letter groups (order does not matter) can be chosen from A, B, C, D?

A. 8

B. 6

C. 4

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5

How many ordered pairs of different letters can be made from A, B, C, D?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

An organized list of groups should include both AB and BA when order does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 1

Count selections from separate groups • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A committee needs 3 of 5 boys and 2 of 4 girls. How many committees are possible?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

How many ways can 2 hearts be chosen from the 13 hearts in a standard 52-card deck?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

True or false?

$nCr = n!/(r!(n - r)!)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

How many 5-card hands can be chosen from a standard 52-card deck (order does not matter)?

A. 2,598,960

B. 380,204,032

C. 260

D. 311,875,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5

True or false?

Choosing 2 from 4 girls and 3 from 5 boys gives 6 + 10 = 16 committees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 2

Turn counts into probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

3 people are chosen at random from 5 boys and 4 girls. How many groups are possible in all?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

3 people are chosen at random, without replacement (every group of 3 equally likely), from 5 boys and 4 girls. What is P(all three are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

True or false?

In a uniform model, P(event) = (number of outcomes in the event)/(number of outcomes in all).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

2 cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both hearts)?

A. 1/17

B. 1/16

C. 1/4

D. 13/52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

P(all three are boys) from 5 boys and 4 girls is 3/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

Compute probabilities with permutations and combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count with permutations and combinations	**HSS-CP.B.9**
BACKWARD 2	Multiply conditional probabilities along draws	**HSS-CP.B.8**
FORWARD 1	Use complements with counting	**HSS-CP.B.9**
FORWARD 2	Define a random variable from a counting model	**HSS-MD.A.1**

SET 45 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A lottery draws 3 different numbers from 1 to 10 (order does not matter). What is the probability that one ticket with 3 numbers wins? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(all girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

Five people line up in random order. What is P(Ann is first and Bo is second)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

Two cards are dealt at random, without replacement, from a standard 52-card deck. What is P(both are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(exactly one girl)?

A. 1/3

B. 5/21

C. 10/21

D. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q5**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 1

Count with permutations and combinations • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Compute 10C3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

Compute 5!.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

True or false?

$9C3 = 84$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

Compute 52C2.

A. 676

B. 1,326

C. 2,652

D. 104

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$5C2 = 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 2

Multiply conditional probabilities along draws • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

Using the Multiplication Rule, find P(both hearts) for two cards dealt at random, without replacement, from a standard 52-card deck. Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

True or false?

$(4/9)(3/8)(2/7) = 1/21$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

From 5 boys and 4 girls, two are chosen in order without replacement. P(both boys)?

A. 25/81

B. 1/4

C. 5/9

D. 5/18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5

Five people line up at random. P(Ann first)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

True or false?

A group has 5 boys and 4 girls. Two people are chosen one after the other, at random and without replacement. Given that the first is a girl, P(second is a girl) = 4/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 1

Use complements with counting • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(no girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(at least one girl)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

True or false?

P(at least one) = 1 − P(none).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

Four cards are dealt at random, without replacement, from a standard 52-card deck. What is P(no aces)? (48C4 = 194,580 and 52C4 = 270,725.)

A. 194,580/52

B. 4/52

C. 1/13

D. 38,916/54,145

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

True or false?

P(at least one girl) = 3 · (4/9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 2

Define a random variable from a counting model • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What values can X take? Give the smallest and the largest.

Smallest: _____

Largest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. Find P(X = 0) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

True or false?

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. The probabilities P(X = 0), P(X = 1), P(X = 2), P(X = 3) add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What is P(X = 2)?

A. 5/14

B. 2/9

C. 1/21

D. 10/21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

True or false?

A random variable assigns a probability, not a number, to each outcome.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

Define a random variable and graph its distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.1**
BACKWARD 1	Describe outcomes in a sample space	**HSS-CP.A.1**
BACKWARD 2	Read and draw a distribution display	**HSS-ID.A.1**
FORWARD 1	Find probabilities of ranges of values	**HSS-MD.A.1**
FORWARD 2	Compute the expected value of a random variable	**HSS-MD.A.2**

SET 46 • HSS-MD.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

Three fair coins are flipped independently. X = number of heads. Find P(X = 0), P(X = 1), P(X = 2) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

Two fair dice are rolled independently and X = the sum. How many different values can X take, and what is P(X = 7)? Give a fraction in lowest terms.

Number of values: _____

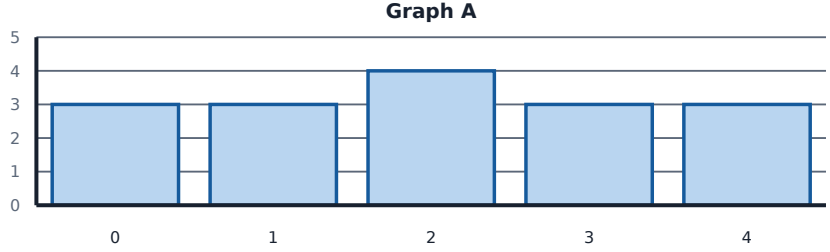
P(X = 7): _____

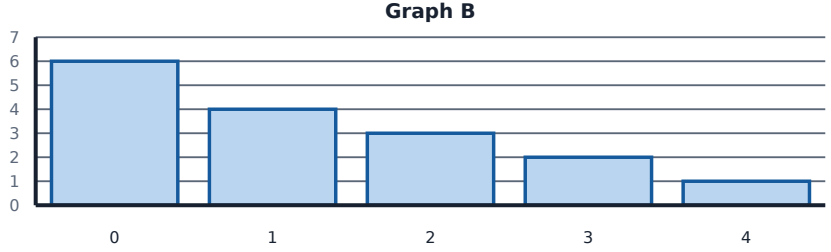
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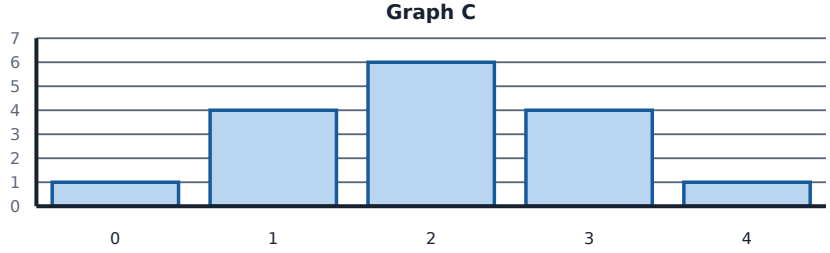
ID S46-M-Q2

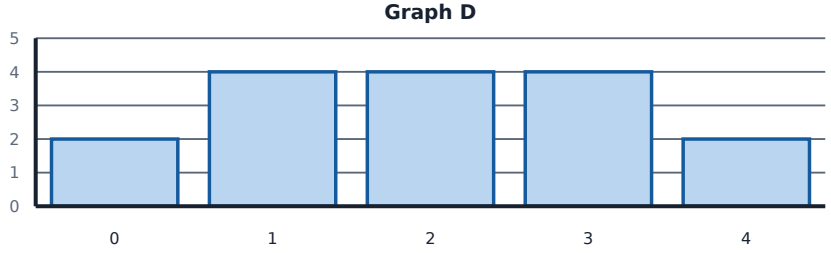
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

Four fair coins are flipped independently and X = number of heads. Each bar graph shows, for X = 0 to 4, how many of the 16 equally likely outcomes give that value. Which graph shows the distribution of X? In each histogram, the horizontal axis shows number of heads and the vertical axis shows frequency.









A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q3/5

Which list could be a probability distribution for X = 1, 2, 3?

A. 0.3, 0.4, 0.4

B. 0.2, 0.2, 0.2

C. 0.5, −0.1, 0.6

D. 0.2, 0.5, 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

X takes the values 0, 1, 2, 3 with P(X = 0) = 0.1, P(X = 1) = 0.35 and P(X = 3) = 0.15. Find P(X = 2).

P(X = 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 1

Describe outcomes in a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Three fair coins are flipped independently. How many outcomes have exactly 1 head?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

True or false?

Three fair coins are flipped independently. The 8 outcomes of the sample space are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q2/5

Two fair dice are rolled independently. How many outcomes have a sum of 4?

A. 3

B. 2

C. 6

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5

Four fair coins are flipped independently. How many outcomes are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

For two fair dice rolled independently, the sums 2 and 12 each come from 2 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 2

Read and draw a distribution display • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

True or false?

In a bar graph of a distribution, taller bars show more likely values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

A bar graph shows counts 1, 4, 6, 4, 1 for X = 0 to 4. What is its shape?

A. skewed right

B. uniform

C. skewed left

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

A bar graph has counts 1, 4, 6, 4, 1 for X = 0 to 4. What is the total count?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

True or false?

A bar graph with counts 6, 4, 3, 2, 1 is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

Counts are 1, 4, 6, 4, 1 for X = 0 to 4. Which value of X is most likely?

A. 1

B. 2

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 1

Find probabilities of ranges of values • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

For X = number of heads in 3 flips, find P(X ≥ 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

For X = sum of two fair dice rolled independently, find P(X ≤ 4). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

True or false?

The probabilities of all values of a random variable add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

X takes the values 0 to 3 with probabilities 0.1, 0.35, 0.4, 0.15. What is P(X > 1)?

A. 0.9

B. 0.45

C. 0.4

D. 0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

True or false?

$P(X \geq 1) = 1 - P(X = 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 2

Compute the expected value of a random variable • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 2 fair coins are flipped independently. Find the expected value of X.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

True or false?

The expected value is the probability-weighted average of the values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

X takes 1, 2, 3 with probabilities 0.2, 0.5, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

X = result of spinning a spinner with 4 equal sections numbered 1 to 4. What is E(X)?

A. 2

B. 4

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5

True or false?

The expected value must be one of the values X can take.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

Calculate and interpret expected value (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.2**
BACKWARD 1	Compute a mean	**6.SP.B.5.c**
BACKWARD 2	Read a probability distribution	**HSS-MD.A.1**
BACKWARD 3	Multiply and add decimals	**6.NS.B.3**
FORWARD 1	Expected value with gains and losses	**HSS-MD.A.2**
FORWARD 2	Find expected values of theoretical distributions	**HSS-MD.A.3**

SET 47 • HSS-MD.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 3 fair coins are flipped independently (table). Find E(X).

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

Find E(X) for the distribution in the table.

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

X = the number rolled on a fair six-sided die. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

For the distribution in the table, E(X) = 1.6. What does this mean?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

A. The single most likely value of X is 1.6.

B. X will equal exactly 1.6 on most of the trials.

C. Over many trials, the average value of X is about 1.6.

D. X is 1.6 or less with a probability of 1.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions

HSS-MD.A.2 • Set 47 • MAIN • Q4/5

A game pays $10 with probability 0.2, $2 with probability 0.3, and $0 otherwise. Find the expected payout.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5, 6.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5

Find the mean of 0, 1, 1, 1, 2, 2, 2, 3.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5

True or false?

A value listed 3 times counts 3 times in the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5

Find the mean of 2, 2, 2, 8.

A. 14

B. 5

C. 2

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5

True or false?

The mean of 0, 0, 0, 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 2

Read a probability distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

In the distribution table with values 0 to 3, what is P(X = 2)?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

True or false?

A probability distribution lists every value of X with its probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

In the table (X = number of heads when 3 fair coins are flipped independently), which values of X are equally likely?

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

A. 2 and 3 **B.** 0 and 1 **C.** all four **D.** 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

A distribution has $P(X = 5) = 0.3$ and $P(X = 10) = 0.7$. What is $P(X = 5) + P(X = 10)$?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A distribution may have probabilities adding to 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 3

Multiply and add decimals • Grade 6 • The Number System • 6.NS.B.3

Compute 2(0.4) + 3(0.15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5

Compute 10(0.2) + 2(0.3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5

True or false?

$1(0.35) + 2(0.4) = 1.15.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5

Compute 5(0.3) + 10(0.7).

A. 8.5

B. 7

C. 1.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5

True or false?

$4(0.25) + 2(0.5) = 1.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 1

Expected value with gains and losses • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

A game: win $5 with probability 0.2, lose $2 with probability 0.8. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

True or false?

A negative expected gain means a player loses money on average over many plays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

X has values −3, 0, 6 with probabilities 0.5, 0.2, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

A game has expected gain $-\$0.60$. About how much does a player expect to lose in 100 plays?

A. $600

B. $6

C. $0.60

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

True or false?

If E(X) = 0, the player breaks even on every single play.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 2

Find expected values of theoretical distributions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 3). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). Find the expected number correct.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

For n independent trials with success probability p, the expected number of successes is np.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

A student guesses at random on 10 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is the expected number correct?

A. 2.5

B. 0.25

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). The probability of getting all 4 right is 1/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

Develop a theoretical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.3**
BACKWARD 1	Count arrangements of successes	**HSS-CP.B.9**
BACKWARD 2	Multiply probabilities of independent trials	**HSS-CP.A.2**
FORWARD 1	Build the whole theoretical distribution	**HSS-MD.A.3**
FORWARD 2	Compare a theoretical distribution with data	**HSS-MD.A.4**

SET 48 • HSS-MD.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

A student guesses at random on 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). Find P(all 5 correct) and the expected number correct. Give each answer as a fraction in lowest terms.

P(all correct): _____

Expected number correct: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(exactly 1 correct)?

A. 27/64

B. 9/64

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(at least 1 correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 1

Count arrangements of successes • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

In how many ways can exactly 2 of 4 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

In how many ways can exactly 1 of 3 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

True or false?

$5C5 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

In how many ways can exactly 3 of 5 answers be correct?

A. 10

B. 60

C. 15

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$4C2 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 2

Multiply probabilities of independent trials • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Compute $(1/2)^4$. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

Compute (1/4)(3/4)(3/4). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

True or false?

When each guess is made at random and independently, being right on one question does not change the chance of being right on another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

P(correct) = 1/4 on each of 2 independent guesses. P(both correct)?

A. 1/8

B. 1/2

C. 1/16

D. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

True or false?

$(3/4)^3 = 9/12$.

A. True

B. False

SET 48 • HSS-MD.A.3

FORWARD 1

Build the whole theoretical distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find P(X = 0), P(X = 1), P(X = 2), P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

True or false?

The four probabilities 27/64, 27/64, 9/64, 1/64 add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Which value of X is least likely?

A. 3

B. 2

C. 0

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5

True or false?

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. X = 1 is more likely than X = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

FORWARD 2

Compare a theoretical distribution with data • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

A fair six-sided die is rolled 60 times, independently. How many sixes does the theoretical distribution predict?

Sixes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

In 60 rolls, a die showed 6 sixes. What is the empirical probability of a six? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

An empirical probability is found from observed data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

In 100 households, the numbers of TVs gave an average of 1.9. This average is

A. a theoretical probability from a model

B. always equal to 2 for any survey

C. the expected value of the empirical distribution

D. a margin of error for the survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5

True or false?

Empirical and theoretical probabilities for the same event are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

Develop an empirical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.4**
BACKWARD 1	Find relative frequencies	**7.SP.C.6**
BACKWARD 2	Compute an expected value	**HSS-MD.A.2**
FORWARD 1	Compare empirical and theoretical results	**HSS-MD.A.4**
FORWARD 2	Find the expected payoff of a game of chance	**HSS-MD.B.5.a**

SET 49 • HSS-MD.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

100 households were surveyed (table). X = number of TVs in a randomly chosen household. Find P(X = 2) and P(X ≥ 3). Give decimals.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

P(X = 2): _____

P(X ≥ 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

Using the TV survey, find the expected number of TVs in a household.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

The 100 surveyed households were a random sample of a town's households. Using the sample mean, 1.9 TVs per household, as the estimate, about how many TVs are expected in 500 households of the town?

TVs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5